Roll No.: BL.AI.U4AID24140

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Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations –March 2024

Second Semester

Artificial Intelligence & Data Science

23AID111 Object Oriented Programming

Duration: Two hours

Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Represent the problems using objects and classes.
CO02	Implement object-oriented concepts using the Java language.
CO03	Apply object-oriented concepts to design and visualize programs using UML.
CO04	Implement applications using object-oriented features.

Q1. Multiple Choice Questions. Each Question Carries 1 Marks . All questions are compulsory.[10 Marks]

a. [1] [CO01][BTL 2]

```
public class Test {
   public static void main(String[] args) {
      int x = 5;
      int y = x++ + ++x;
      System.out.println(y);
   }
        }
```

What will be the output?

i) 10

ii) 12

iii) 11

iv) 13

b [1] [CO01][BTL 2]

```
public class Test {
   public static void main(String[] args) {
      double d = 10 / 4;
      System.out.println(d);
   }
}
```

What will be the output?

i) 2.5

ii) 2.0

iii) 2
iv) Compilation error

c
```
class Test {
    public static void main(String args[]) {
        final int x = 10;
        x = 20;
        System.out.println(x);
    }
}
```
[1] [CO01] [BTL 2]

What will be the output?

i) 10
ii) 20
iii) Compilation error
iv) Runtime error

d
```
class Test {
    void display(int a) {
        System.out.println("Integer: " + a);
    }
    void display(double a) {
        System.out.println("Double: " + a);
    }
    public static void main(String[] args) {
        Test obj = new Test();
        obj.display(5);
    }
}
```
[1] [CO01] [BTL 2]

What will be the output?

i) Integer: 5
ii) Double: 5.0
iii) Compilation error
iv) Ambiguity error

e
```
class Parent {
    void show() {
        System.out.println("Parent class");
    }
}
class Child extends Parent {
    void show() {
        System.out.println("Child class");
    }
}
public class Test {
    public static void main(String[] args) {
        Parent obj = new Child();
        obj.show();
```
[1] [CO01] [BTL 2]

```
    }
}
```

What will be the output?

i) Parent class
ii) Child class
iii) Compilation error
iv) Runtime error

f

```
interface A {
    void show();
}
class B implements A {
    public void show() {
        System.out.println("Hello from B");
    }
}
public class Test {
    public static void main(String[] args) {
        A obj = new B();
        obj.show();
    }
}
```

[1] [CO01] [BTL 2]

What will be the output?

i) Hello from B
ii) Compilation error
iii) Interface cannot be instantiated
iv) Runtime error

g

```
class Parent {
    Parent() {
        System.out.println("Parent Constructor");
    }
}
class Child extends Parent {
    Child() {
        System.out.println("Child Constructor");
    }
}
public class Test {
    public static void main(String[] args) {
        Child obj = new Child();
    }
}
```

[1] [CO01] [BTL 2]

What will be the output?

i) Parent Constructor
Child Constructor
ii) Child Constructor

Parent Constructor

iii) Compilation error

iv) Runtime error

h

```
class A {
   final void show() {
      System.out.println("A's show");
   }
}
class B extends A {
   void show() {
      System.out.println("B's show");
   }
}
public class Test {
   public static void main(String[] args) {
      B obj = new B();
      obj.show();
   }
}
```

[1] [CO01] [BTL 2]

What will happen?

i) A's show

ii) B's show

iii) Compilation error

iv) Runtime error

i

```
class A {

   A() {
      System.out.println("Constructor of A");
   }
}

class B extends A {
   B() {
      System.out.println("Constructor of B");
   }
}

class C extends B {
   C() {
      System.out.println("Constructor of C");
   }
}

public class Test {
   public static void main(String[] args) {
```

[1] [CO01] [BTL 2]

```
        C obj = new C();
    }
}
```

What will be the output?

i) Constructor of C
   Constructor of B
   Constructor of A

ii) Constructor of A
   Constructor of B
   Constructor of C

iii) Compilation error

iv) Runtime error

j

```
public class Test {

    public static void main(String[] args) {
        int a = 5;
        int b = a++ + ++a;
        System.out.println(b);
    }
}
```

[1] [CO01] [BTL 2]

What will be the output?

i) 10
ii) 11
iii) 12
iv) 13

Q2. Short Answer Questions. Each Question carries 5 Marks.Attempt any four

a. Explain the concept of Encapsulation in Java. How does it help in data hiding? Provide an example. [5] [CO01] [BTL 1]

b. Explain method overriding in Java with an example program [5] [CO02] [BTL 2]

c. Describe the importance of constructors in Java. How do default and parameterized constructors differ? [5] [CO01] [BTL 1]

d. Write a Java program to demonstrate the use of the final keyword with variables and methods. [5] [CO02] [BTL 2]

e. What is polymorphism in Java? How does method overloading achieve compile-time polymorphism? [5] [CO01] [BTL 2]

f. What are jagged arrays in Java? Write a program to illustrate their use. ? [5] [CO02] [BTL 2]

Q3. Write a program in java and also write the output. Each Question carries 10 Marks. Attempt any two. [10] [CO01] [BTL 2]

a. Write a Java program to design an Employee Management System. It should have:
- A base class Employee with attributes name and salary.
- A derived class Manager that overrides the display() method and has an extra attribute department.
- Demonstrate method overriding, inheritance, and constructors.

b. Create a Java program demonstrating single inheritance using the following classes:
- A base class Vehicle with an attribute brand (String) and a method showBrand() to display the brand.
- A derived class Car that extends Vehicle, adds an attribute model (String), and a method showDetails() to display the brand and model.
- In the main method, create an object of Car, set values, and display details.

c Design a Java program that demonstrates the use of interfaces.
1. Create an interface Vehicle with the following methods:
   - void start()
   - void stop()
2. Implement this interface in two classes:
   - Car: Implements start() by printing "Car is starting" and stop() by printing "Car is stopping".
   - Bike: Implements start() by printing "Bike is starting" and stop() by printing "Bike is stopping".

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**20**	BTL 1	**20**
CO02	**30**	BTL 2	**30**